Sorting Techniques: Bubble Sort, Insertion Sort & Selection Sort Data Structures

> **Definition: Sorting** is the algorithmic process of arranging elements of a collection into a systematic order (ascending or descending).

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1. Detailed Technical Explanation

1. Bubble Sort (Comparison & Swap)

- **Mechanism:** Repeatedly steps through the list, compares adjacent elements, and swaps them if they are in the wrong order. The largest element "bubbles up" to its correct position at the end of each pass.

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### 2. Selection Sort (Minimum Element Selection)

- **Mechanism:** Divides array into sorted and unsorted subarrays. Repeatedly finds the **minimum element** from the unsorted subarray and places it at the beginning of the unsorted part.

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### 3. Insertion Sort (Card Player's Sorting)

- **Mechanism:** Builds the sorted array one item at a time by picking the next element and inserting it into its correct relative position within the already-sorted left subarray.

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## 2. Comprehensive Comparison Table

| Algorithm | Best Time | Average Time | Worst Time | Space | Stable? | In-Place? |

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## 3. Quick Recall Flow

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Bubble